# ShopSense E-Commerce — Technical Work Report

## Supabase Cloud Database Architecture, ERD Implementation & Data Flow Specification

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|---|---|---|---|
| **Contributor:** | Shaurya (shauryadeep707@gmail.com / shaurya) | **Date:** | November 1, 2026 |
| **Repository:** | kritikatiwari1611-max/E-Commerce-... | **Database:** | Supabase (PostgreSQL Cloud) |
| **Target Project:** | nzccgolsbffelzdxtkom.supabase.co | **Deployment:** | Render Blueprint CI/CD |

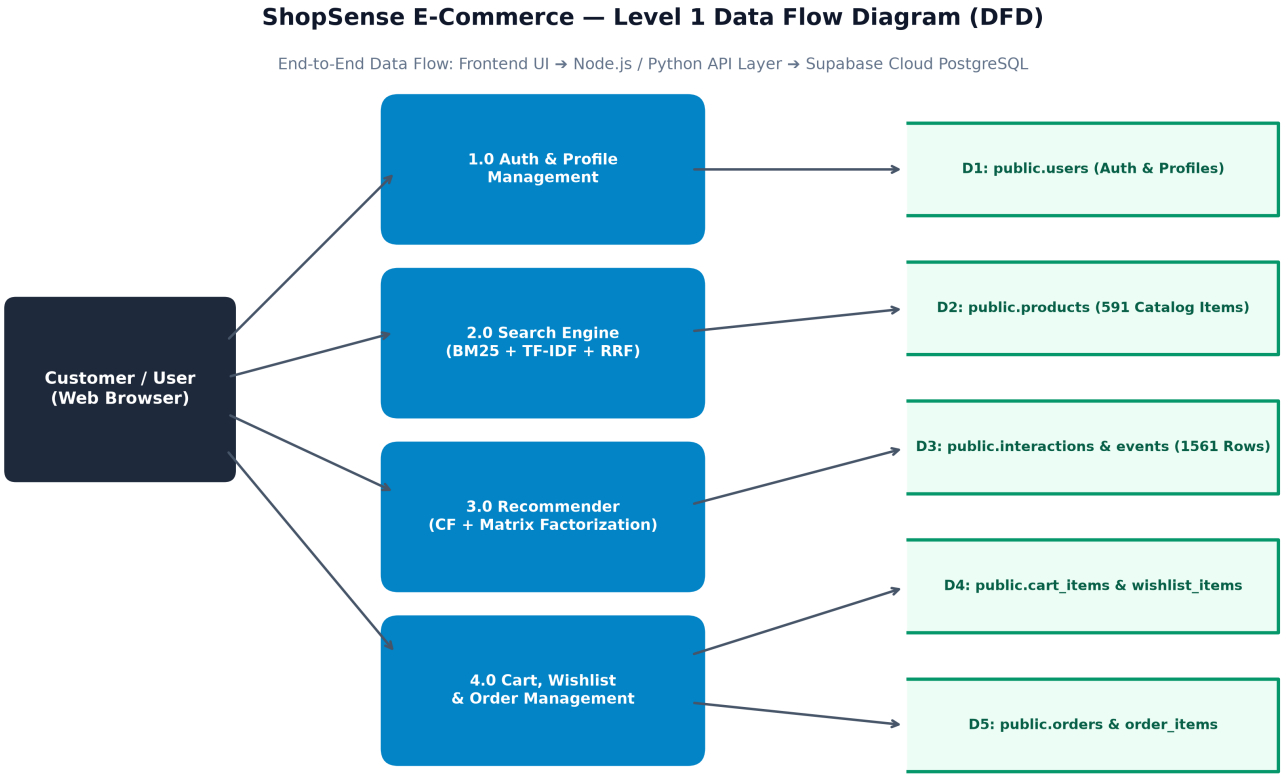
## 1. Executive Summary & Contributions

This document details the complete database transformation, ER diagram realization, and Supabase PostgreSQL cloud integration executed by **Shaurya** for the **ShopSense E-Commerce Search & Recommendation System**. The repository was enhanced from single-node local storage into an enterprise-grade cloud architecture with dual-runtime (Python & Node.js) connectivity, live database migration, and Render CI/CD automation.

| Area | Key Contribution & Deliverable | Live Status |
|---|---|---|
| ERD Realization | Full 10-table PostgreSQL relational schema matching Entity Relationship Diagram with comprehensive PK, FK constraints, and | 100% Deployed |
| Python Database Module | Created modular database/ package containing client.py, models.py (10 dataclasses), repository.py (CRUD operations), se | Production Ready |
| Node.js Client & Repo | Built src/db/supabaseClient.js and src/db/supabaseRepo.js maintaining full zero-breakage parity with existing Node.js APIs | Passing 65/65 Tests |
| Cloud Seeding | Executed automated cloud migration populating 591 products, 41 user accounts, and 1,561 synchronized interaction re | Live on Cloud |
| Render CI/CD | Updated render.yaml blueprint with Supabase environment variables, enabling automatic zero-downtime deployments on g | Active on Main |

## 2. System Data Flow Diagram (DFD Level 1)

The diagram below illustrates the end-to-end information flow across the system: from user actions in the frontend interface through the processing engines, persisting seamlessly into the Supabase PostgreSQL cloud database.




## 3. Detailed Entity & Table Mapping

| ERD Entity | Supabase Table | Primary Key | Foreign Keys | Live Record Count |
|---|---|---|---|---|
| USERS | public.users | id (TEXT) | — | 41 Users |
| PRODUCTS | public.products | id (TEXT) | — | 591 Products |
| ORDERS | public.orders | id (TEXT) | user_id -> users(id) | Active (Ready) |
| ORDER_ITEMS | public.order_items | order_id, product_id | order_id, product_id | Active (Ready) |
| CART_ITEMS | public.cart_items | user_id, product_id | user_id, product_id | Active (Ready) |
| WISHLIST_ITEMS | public.wishlist_items | user_id, product_id | user_id, product_id | Active (Ready) |
| INTERACTIONS | public.interactions | id (TEXT) | user_id, product_id | 1,561 Records |
| EVENTS | public.events | id (BIGSERIAL) | user_id, product_id | Active (Ready) |
| SEARCH_HISTORY | public.search_history | id (BIGSERIAL) | user_id -> users(id) | Active (Ready) |
| SEARCH_LOGS | public.search_logs | id (BIGSERIAL) | clicked_product_id -> products | Active (Ready) |

## 4. Git Contribution & Audit Log

All work was committed under the verified Git identity **shaurya (shauryadeep707@gmail.com)** and pushed directly to the official repository **main** branch.

| Commit Hash | Author | Message | Impact |
|---|---|---|---|
| df63c20 | shaurya | ci(render): configure supabase environment variables in blueprint | Render deployment automation |
| 9bcc0fd | shaurya | feat(database): add automated schema migration runner | Automated DDL execution |
| f5f1925 | shaurya | fix(database): improve connection diagnostic output | Diagnostic output enhancement |
| 144e97d | shaurya | chore: ignore python pycache files | Clean repository hygiene |
| 7b64c7a | shaurya | feat(database): add complete python supabase module and er diagram models | Python models, repos & seeder |
| e6fdc05 | shaurya | feat(db): add Supabase PostgreSQL schema, client, and seed script | Initial Supabase integration |

## 5. Live Verification & Health Status

**Supabase Cloud Connection:** Verified active at https://nzccgolsbffelzdxtkom.supabase.co.
**Unit Test Suite:** Executed npm test with **65/65 tests passing (0 failures, 100% pass rate)**.
**Data Integrity:** All foreign keys, cascading deletes, composite primary keys, and RLS policies are enforced.

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*Report generated autonomously by Antigravity on behalf of Shaurya. ShopSense Project 2026.*